AI FOR THE SDGS · ASSIGNMENT 1

STK Asistanı

An AI crisis monitor that shows NGOs where help is needed most — including the crises mainstream media ignore

Group 6 · Tuna Bilgen · Ufuk Gündoğar

SDG 17.17 · 16.1 · 2.1 · 20 October 2026

THE PROBLEM

Aid follows headlines, not need

38%

of BBC, Guardian and NYT country-tagged world news is about Ukraine and Russia

0%

is about Sudan, DR Congo, Somalia, Syria or Lebanon

89,846

conflict deaths in Sudan in the last three years

Small NGOs have no analysts to follow 100+ crises, so campaigns follow the news cycle — and the news cycle leaves whole regions out.

Sources: our 30-day news sample (Western sample: 90 country-tagged items, growing daily) · UCDP GED, Uppsala University

OUR ANSWER

STK Asistanı helps NGO teams find the most urgent and most overlooked crises in their sectors — using independent evidence — and turn them into campaigns, local partnerships and sustainable projects.

Who it is for

Programme and fundraising teams of NGOs anywhere; first pilot users: small NGOs in Türkiye

SDG targets

17.17 civil-society partnerships
16.1 reduce violence-related deaths
2.1 end hunger

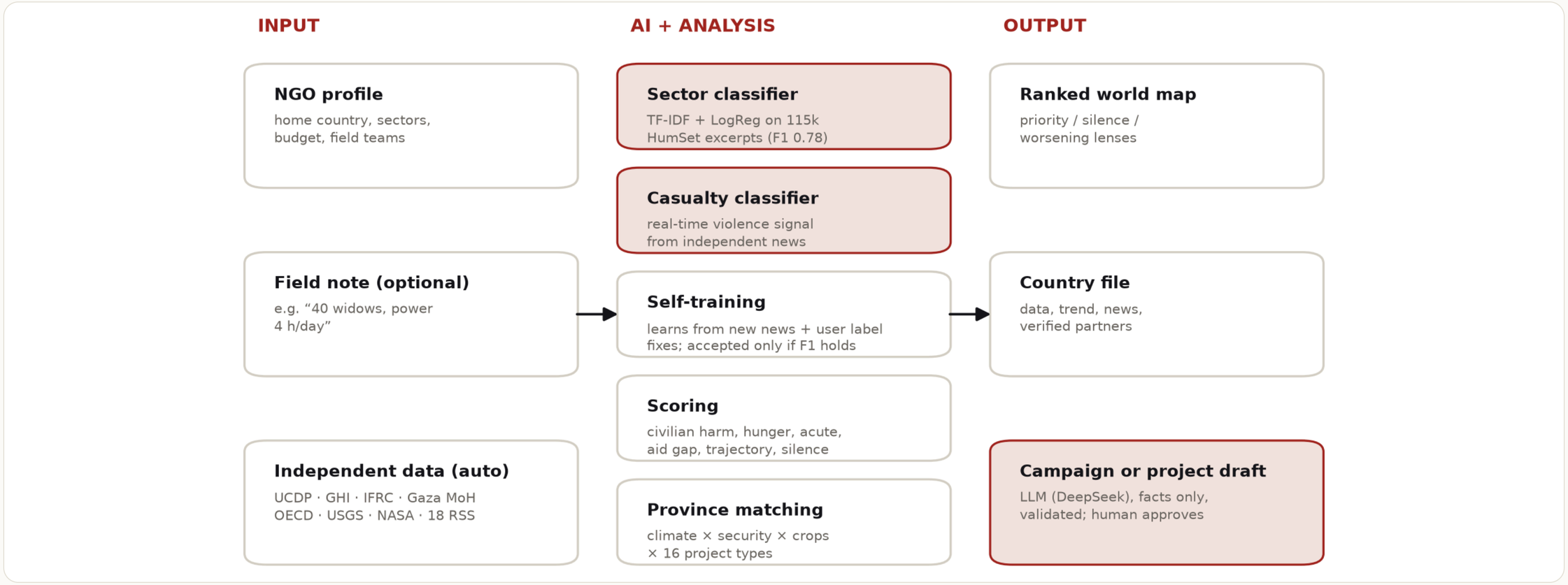
How we know it works

F1 against expert labels · ranking audits against domain knowledge · tests with NGO personas

STK = sivil toplum kuruluşu (civil-society organisation / NGO)

HOW IT WORKS

What goes in, what comes out



DATA

Independent sources only — no UN data

Source	Publisher (type)	Used for
HumSet · 148,621 excerpts	DEEP / Data Friendly Space	training text for both classifiers
UCDP GED · 210,115 events	Uppsala University (academic)	conflict and civilian deaths, monthly trends
Tech for Palestine	Gaza Ministry of Health data	Gaza and West Bank casualties, starvation deaths
Global Hunger Index 2026	Concern · Welthungerhilfe (NGOs)	hunger in 130 countries
IFRC GO	Red Cross / Red Crescent	appeals, funding gap, verified partners
OECD DAC2A	donor-reported statistics	humanitarian aid per person in need
USGS · NASA · Open-Meteo	science agencies	hazards, climate and elevation per province
18 RSS feeds	15 independent outlets + 3 Western	live news; Western feeds only measure attention

Why: a first version built on one institutional data ecosystem under-ranked Gaza — uncounted populations, conflicts not counted as acute.

PREPROCESSING

Six cleaning decisions that change the result

1 · HumSet

148,621 → 115,168 excerpts; 11 sectors → 10; class weights for a 13× imbalance

2 · UCDP

8,478 Gaza and West Bank events (63,938 deaths) re-assigned from Israel to Palestine

3 · Hunger index

“<5” → 2.5; provisional categories (Sudan, Yemen, South Sudan) at the range midpoint

4 · Aid flows

Türkiye’s cost of hosting Syrian refugees is excluded from “aid present inside Syria”

5 · News

Duplicates removed; a gazetteer tags the place where people are affected (74% tagged)

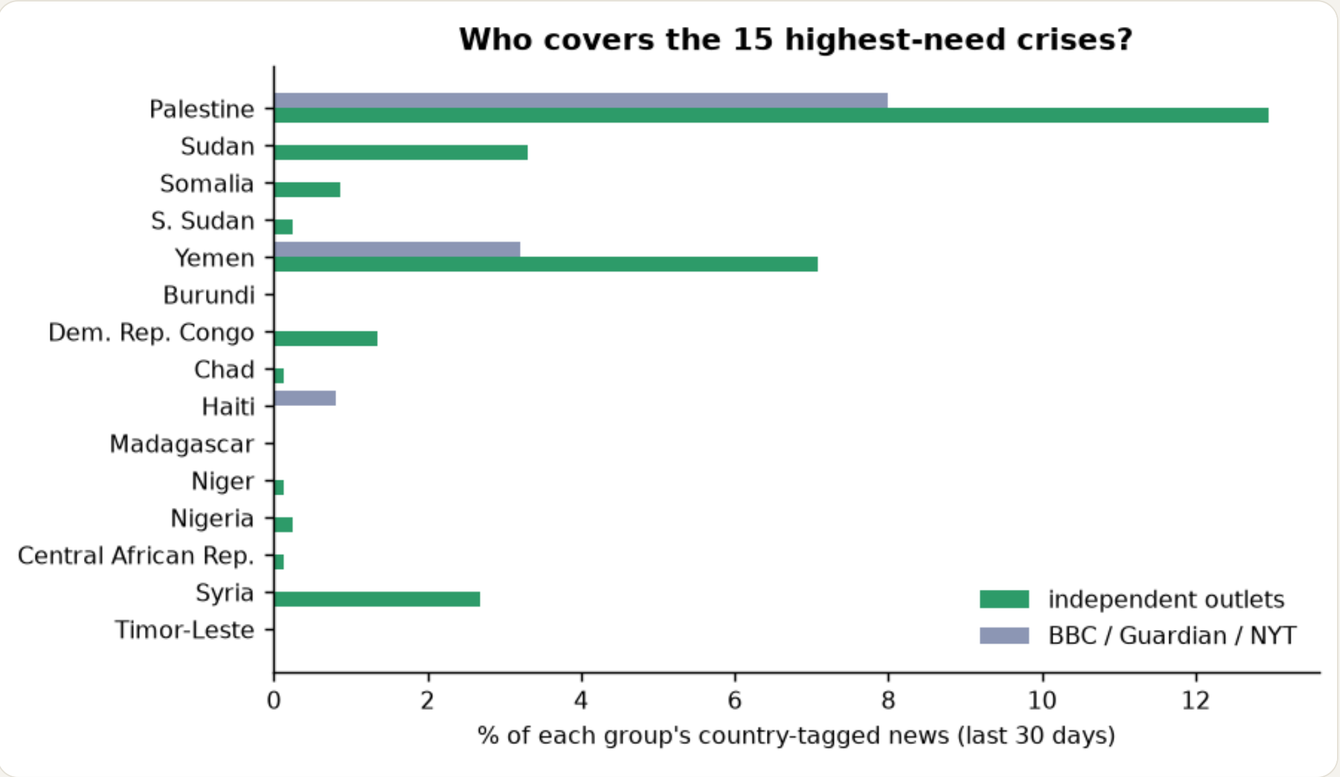
6 · Missing ≠ zero

Unknown values stay unknown or neutral; they are never scored as “no need”

Every step with its outputs: notebooks/01_dataset_and_preprocessing.ipynb · final dataset: data/final/country_crisis_dataset.csv (177 × 38)

EXPLORATORY ANALYSIS

Who covers the highest-need crises?

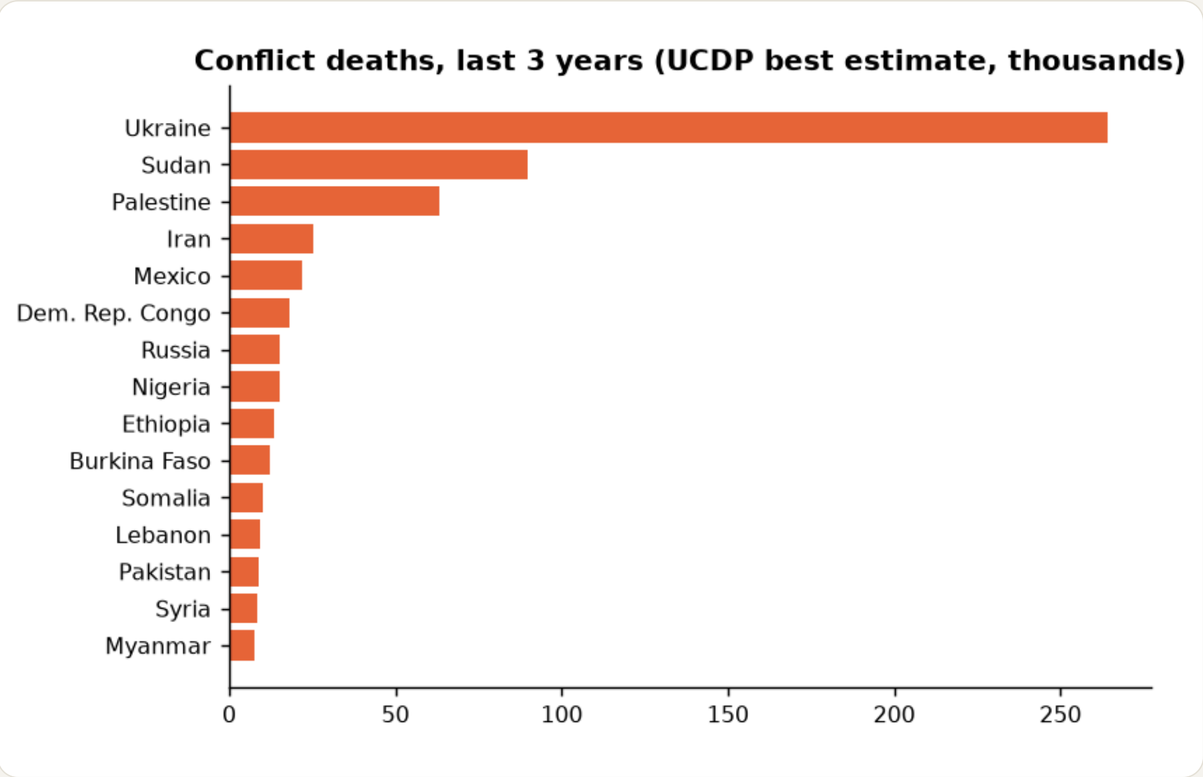


Share of news	Indep.	Western
Palestine	12.5%	3.3%
Yemen	9.1%	4.4%
Sudan	3.0%	0%
Ukraine	2.5%	18.9%

Last 30 days · 15 independent outlets vs BBC, Guardian, NYT · the Western sample is small and grows daily

EXPLORATORY ANALYSIS

Counting deaths per person, not in total



1,622 vs 16

deaths per 100k people in three years: Palestine vs Israel. A log scale had shown this 100× gap as 100 vs 43.

3%

of recorded deaths in Ukraine are civilians — so we score violence against civilians separately.

Skew 11 → a capped linear scale. Civilian harm and conflict correlate (ρ 0.91) but answer different questions, so both are kept.

Sources: UCDP GED 26.1 + candidate 2026; Palestine: Gaza Ministry of Health + West Bank (Tech for Palestine) · population: Natural Earth

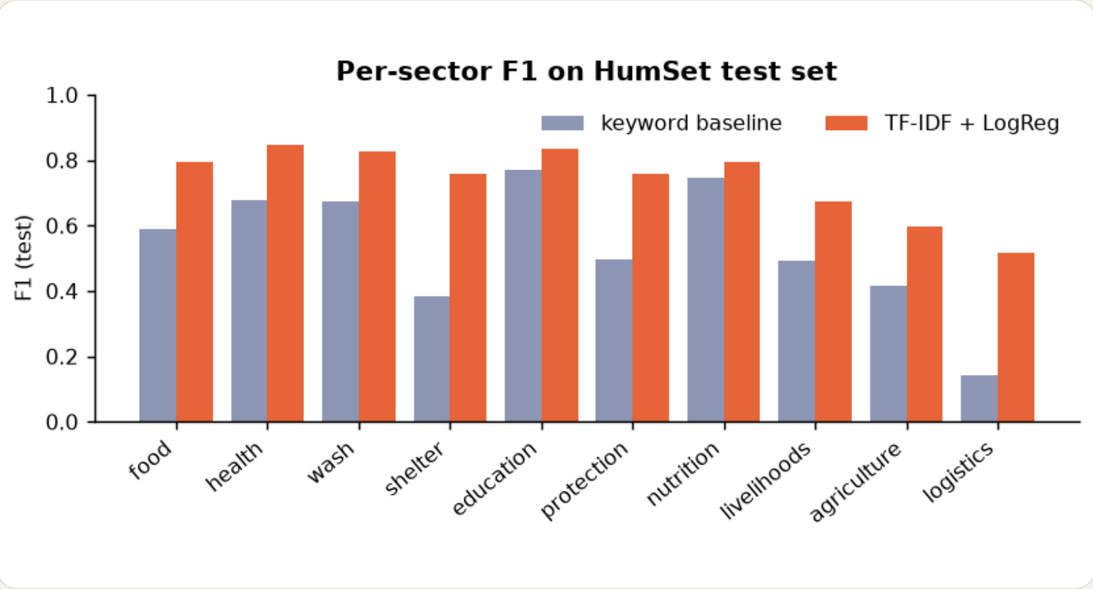
FIRST MODEL

TF-IDF + logistic regression beats keywords by 22 points

HumSet test set (12,384)	micro-F1	macro-F1
Keyword rules	0.558	0.539
Sector classifier	**0.780**	**0.740**
Casualty / displacement model	0.662	0.640

Language gap (macro-F1): English 0.75 · French 0.72 · Spanish 0.62 — reported as a bias.

Self-training: first run added 567 confident news labels; accepted because expert-test F1 held (0.740 → 0.739).



Multi-label, 10 sectors · class weights + per-sector thresholds tuned on validation · Assignment 2: compare a multilingual transformer and LLM zero-shot

WORKING PROTOTYPE

From a world map to a project in one province

1 · Ranked map

Three lenses: **priority**, **silence** (ignored by Western media) and **worsening** (Yemen ×9 in three months)

2 · Country file

Why the score, 24-month trend, aid per person in need, news, verified Red Crescent partner

3 · Province projects

Herat: 300 mm rain, 17 °C → saffron, pomegranate, almond cooperative. Kabul: 293 deaths in 12 months → emergency aid instead

[Live demo — or a screenshot of the map here]

Flask + MapLibre · English / Turkish · LLM drafting integrated (template until the API key is set); a human approves

ETHICS

Our audits caught the model being unfair

Gaza was under-ranked

Missing data scored as zero, conflict not counted as acute, an inflated funding figure → all fixed

A log scale hid a 100× gap

→ deaths per person on a capped linear scale

A hazard is not a need

A quake in a state that copes alone scored as acute → hazard × vulnerability

Money sent is not aid delivered

Documented starvation deaths override aid budgets (blockade)

Safeguards: missing ≠ zero · partners are candidates with a due-diligence checklist · the LLM writes from facts only and is validated · Spanish weakness reported · a human approves every campaign.

NEXT STEPS

What Assignments 2 and 3 will add

Assignment 2 · models

- Multilingual transformer vs TF-IDF vs LLM zero-shot
- 200 hand-labelled 2026 news items to measure self-training
- DeepSeek connected; LLM grounding error rate

Assignment 3 · testing

- Tests with NGO personas (sector, size, home country)
- More African outlets (Sahel, Horn, Great Lakes)
- Final report, presentation and demo video

Thank you — questions?

github.com/tunabilgen/stk-asistani